

BaroDepth: A Method of Estimating Depth with Barometric Pressure Sensors in Smartphones

Eiji Kumakawa*
NTT DOCOMO, INC.
Tokyo, Japan
eiji.kumakawa.ef@nttdocomo.com

Wataru Yamada
NTT DOCOMO, INC.
Tokyo, Japan
wataruyamada@acm.org

Keiichi Ochiai
NTT DOCOMO, INC.
Tokyo, Japan
keiichi.ochiai@acm.org

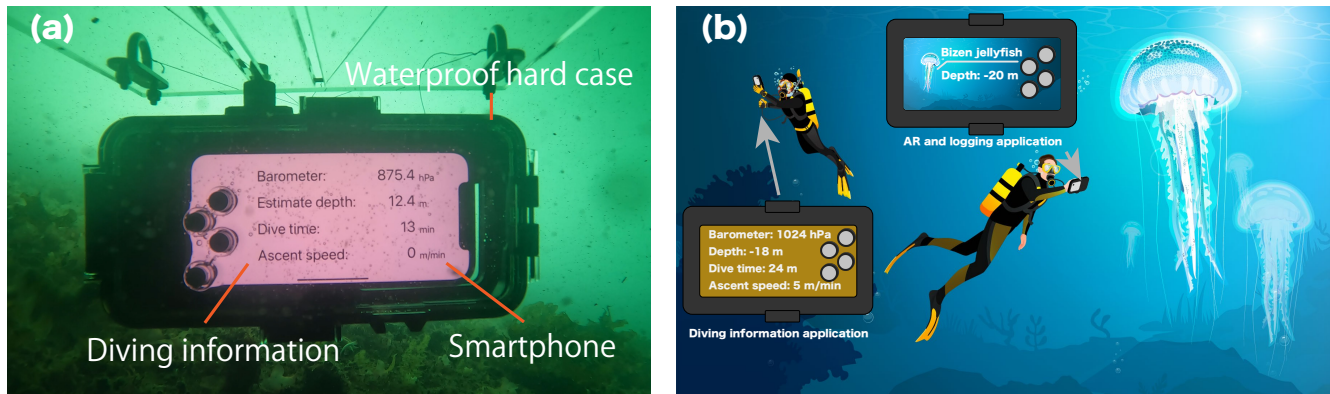


Figure 1: (a) Water depth estimated from changes in the internal pressure of a waterproof case deflected by water pressure. (b) Possible application of the proposed method for underwater activities.

ABSTRACT

Underwater activities are enjoyed around the world. Divers wear dive computers that allow visualization of diving information including depth and dive time, to avoid the risk of decompression sickness. However, dive computers are so expensive that they burden novice divers. A method of estimating water depth with barometric pressure sensors in smartphones, instead of dive computers, was studied [3]. Unfortunately, the conventional method [3] for estimating water depth using a barometric pressure sensor in a smartphone is limited to estimating shallow depths of about 1 m to 2 m due to the sensor only being able to detect up to about 1100 hPa to 1200 hPa. We propose a method of estimating water depths of several tens of meters with a barometric pressure sensor in a smartphone by adopting a waterproof hard case with a decompression adjustment function and a polynomial regression model. We implemented a prototype of the proposed method and conducted experiments to evaluate the precision of estimating water depths down to 20 m in the sea. The experiments revealed that the proposed method could cover water from 0 to 20 m depths and estimate water depths with a root mean squared error (RMSE) of less than 1 m.

*Eiji Kumakawa currently belongs to NTT Social Informatics Laboratories, Kanagawa, Japan (eiji.kumakawa@ntt.com).

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CCS CONCEPTS

• **Human-centered computing** → *Smartphones*; • **Computing methodologies** → *Learning linear models*.

KEYWORDS

water depth estimation, barometer, smartphone

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1 INTRODUCTION

Recreational underwater activities such as diving and snorkeling have become popular worldwide. These underwater activities carry the risk of decompression sickness (DCS), a disease caused by a sudden change in pressure on the body due to water pressure [2]; hence, divers must monitor their water depth, dive time, and ascent rate during diving.

Thus, divers wear a wristwatch-type device called a dive computer that can display and store such information. The information stored and displayed by the dive computer not only ensures safety during diving but also helps plan future dives; however, dive computers are so expensive that they burden novice divers.

Macchia studied a method to estimate water depth using the barometric pressure sensor values in a smartphone equipped with a waterproof soft case that allows smartphones [3]. This method

estimates water depth by detecting changes in air pressure inside the waterproof soft case caused by water pressure pressing on it using a barometric pressure sensor.

Unfortunately, taking into consideration the measurable range of the barometric pressure sensor in a normal smartphone about 1100 hPa to 1200 hPa as discussed in Section 4, the method only works at a depth of 1 m to 3 m; hence, it is impractical for underwater activities such as diving that reach to several tens of meters.

We propose a method to estimate water depth at several tens of meters using a smartphone equipped with a waterproof hard case with an air valve and a polynomial regression model. The proposed method mainly consists of three main ideas. The first idea is to use a waterproof hard case. When a waterproof case for a smartphone is made of soft material, as in the conventional method [3], water pressure is transmitted to the air inside the case with minimal attenuation. As a result, the internal pressure of the case rapidly increases and reaches the upper limit of the barometric pressure sensor even at shallow depths.

A waterproof case made of hard material may not appear to be deformed, but in fact, it becomes slightly deformed underwater due to the effects of water pressure; however, the degree of deformation of a hard case underwater pressure is much lower than that of a soft case. Therefore, the internal air pressure in a waterproof hard case gradually increases and does not reach the upper limit of the barometric pressure sensor even at depths of several tens of meters.

The second idea is to decompress the internal pressure of the waterproof hard case to expand the sensing range of the barometric pressure sensor. If the atmospheric pressure in fine weather is 1000 hPa, the barometric pressure sensor in a smartphone only has a measurement range of only a few hundred hectopascals to the upper limit; hence, smartphones with low upper limits for barometric pressure sensors may not be able to detect sufficient depth. Thus, we expanded the sensing range by reducing the internal pressure of the waterproof case, which serves as the standard for measuring water depth.

The third idea is to apply a polynomial regression model to estimate water depth from the internal pressure of a waterproof hard case, unlike the conventional method that uses a linear regression model and a soft case [3]. Although water pressure and the maximum deflection of a long plate have a proportional relationship, as shown in Section 2.2, the decrease in case volume due to deflection is nonlinear. In addition, since the waterproof hard case normally has a more complex shape and materials than the soft case, it becomes more nonlinear. Therefore, we applied a polynomial regression model built with calibration data to estimate the water depth.

We believe that the proposed method that works at depths of underwater activities will contribute to the development of human-computer interaction underwater. Dive computers are generally expensive pieces of equipment, costing several hundred dollars or more. However, the proposed method can provide diving information such as water depth and ascent speed using only a waterproof case and a daily use smartphone, as shown in Figure 1a and b. This makes it a low-cost alternative to a dive computer for beginner divers.

Second, the proposed method enables the development of smartphone applications that utilize water depth information. Normal

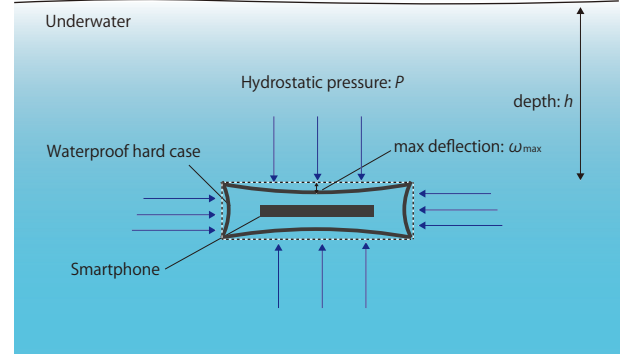


Figure 2: The relationship between water depth and water pressure, and the deflection of the waterproof hard case due to water pressure.

radio communication is not possible underwater because water absorbs radio waves. Thus, unless special means such as wired or optical communication are used, a dive computer cannot communicate with a smartphone.

As shown in Figure 1b, we estimate that the proposed method will enable the development of a variety of applications, not just those that can replace dive computers. Figure 1b shows an augmented reality (AR) application as an example of a possible application. This application not only recognizes the object being photographed by the smartphone but also automatically records its depth. Up to now, users have had to manually record and integrate information from dive computers and smartphones. Therefore, the proposed method supports the development of new smartphone applications for underwater use.

2 PROPOSAL METHOD

We describe the relationship between water depth and water pressure, as well as how the waterproof case deflects due to water pressure.

2.1 Hydrostatic pressure

Hydrostatic pressure is the pressure exerted by the weight of a static fluid as shown in the Figure 2. In an environment with small waves, water pressure can be considered as hydrostatic pressure on the waterproof case [1]. Hydrostatic pressure P_h can be expressed using liquid density ρ , depth from the liquid surface h , and acceleration of gravity g as shown in Equation 1.

$$P_h = \rho gh \quad (1)$$

In other words, Equation 1 indicates that the hydrostatic pressure increases linearly with water depth.

2.2 Deflection of waterproof hard case

Since waterproof hard cases have a slightly deflected surface due to water pressure, as shown in Figure 2, we describe the relationship between the pressure applied to a rectangular plate and the amount of deflection.



Figure 3: Three steps of the proposed method. (a) A smartphone is enclosed in a waterproof hard case with an air valve. (b) A vacuum pump is connected to the air valve to lower the internal pressure of the waterproof case. (c) The depth estimation is calibrated.

The maximum deflection ω_{max} can be expressed using plate thickness t , Young's modulus E , pressure P , and deflection coefficient α , as shown in Equation 2.

$$\omega_{max} = \alpha * P * x^4 / E * t^3 \quad (2)$$

The deflection coefficient α can be uniquely determined from the aspect ratio of the short side a and long side b of a rectangular plate as described in Osadebe et al. [4]. Thus, since the maximum deflection is proportional to the pressure, the maximum deflection increases linearly as water depth increases. In other words, water pressure presses the waterproof hard case and increases its internal air pressure.

2.3 Depth estimation

As shown in Figure 3, the proposed method consisted of three steps. The amount of decompression required in the second step depends on the deflection coefficient of the waterproof case and the measurement range of the barometric pressure sensor. Since the amount of required depressurization is not clear the first time, the internal pressure of the waterproof case is reduced to about 800 hPa or less.

The third step was to measure the internal pressure of the waterproof case at each depth while the smartphone was submerged several meters for calibration. Calibration at deeper depths improves estimation accuracy. Then, a regression equation is derived to estimate the amount of change in the internal pressure of the waterproof case with respect to deeper water depths. A polynomial regression model was adapted to derive the regression equation since some waterproof hard cases have the potential for a nonlinear increase in deflection magnitude caused by water pressure. It is then necessary to check that the estimated internal pressure of the waterproof case at the target depth does not exceed the measurement range of the barometric pressure sensor. If the internal pressure of the waterproof case is expected to exceed the measurement range of the barometric pressure sensor, the case's internal pressure must be further reduced by returning to the second step. Calibration is complete when the estimated internal pressure of the waterproof case at the target depth does not exceed the measurement range of the barometric pressure sensor.

3 EVALUATION

3.1 Water depth estimation experimental conditions

The devices prepared for evaluation experiments are a smartphone (Apple, iPhone 12), a waterproof hard case with an air valve (QSK-01, Temdan) as shown in Figure 3, a pump as an accessory of the waterproof case, and a waterproof soft case (U-FIX, AESOON) for comparison. We experimented in the sea at a depth of about 20 m to 30 m. Environmental conditions on the day of the experiment were clear skies, 20 °C temperature, wind speed of 3 m/s, atmospheric pressure of 1010 hPa at 0m above sea level, and wave height of less than 1 m. The experiment consisted of calibrating depth estimation in shallow water and evaluating barometric pressure sensor values and depth estimation for depths up to 20 m.

3.2 Water depth estimation experiment

The experiment compared three methods of water depth estimation: the waterproof soft case, the waterproof hard case (without decompression), and the waterproof hard case (decompression to 800 hPa).

In the calibration process, the atmospheric pressure value at 0 m above sea level and the internal pressure of the waterproof case at each depth from 1 m to 10 m were acquired, and the regression equation that outputs the estimated water depth was derived from these data. The depth estimation error of each method was evaluated by using root mean square error (RMSE) of the measured and estimated depths taken at 1 m intervals from 1 m to 20 m.

3.3 Result

The results of the experiment consist of two parts: the relationship between the measured water depth and the barometric pressure sensor value, and the error in the water depth estimate.

First, the relationship between the actual measured values of water depth and the air pressure acquired by the smartphone for each method is shown in Figure 4. The method of estimating water depth with the waterproof soft case reached the upper limit of the barometric pressure sensor measurement range at a depth of 3 m, as shown in Figure 4a. This result is because the water pressure at

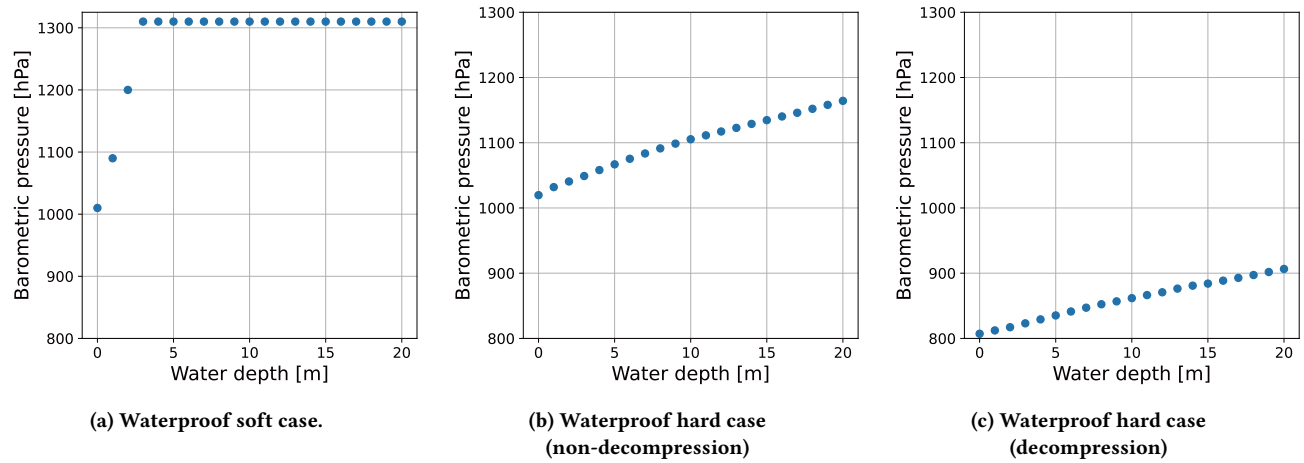


Figure 4: Relationship between water depth and barometric pressure sensor.

a depth of 3 m is about 300 hPa, and the internal pressure of the waterproof soft case that flexibly deforms due to water pressure is directly affected by water pressure. Therefore, a smartphone in a waterproof soft case is not practical to use to estimate depth in underwater activities. On the other hand, both methods using the waterproof hard case revealed changes in the barometric pressure sensor values up to a depth of 20 m regardless of the decompression. This result was attributed to the slight change in the internal pressure of the waterproof hard case, which is less deformed by water pressure than the waterproof soft case. Hence, both methods of using a waterproof hard case were shown to be able to estimate water depth under the conditions of the current experiment.

Second, we explain the results of the evaluation of the error between the estimated depths by the regression model and the measured depths. The results of evaluations performed every 1 m to a depth of 20 m showed RMSE = 0.52 for the waterproof hard case (non-decompression), and RMSE = 0.93 for the waterproof hard case (decompression to 800 hPa), both with RMSE below 1 m. On the other hand, the RMSE of the method using a waterproof hard case with decompression was nearly twice that of the non-decompression method. This result is attributed to the smaller rate of change in barometric pressure sensor values of the method using a waterproof hard case with decompression compared with the non-decompression method, as shown in Figure 4b and c. This is because of the increase in barometric pressure relative to the decrease in volume inside the waterproof hard case due to the deflection of the case being small with the decompression method.

4 DISCUSSION

We have shown that using a waterproof hard case with reduced internal pressure can extend the range of depth estimates. This proposed method is effective even when using a smartphone with a low upper limit of the measurement range of the barometric pressure sensor. For example, an Xperia (SO-02J, Sony Corp.) is equipped with a barometric pressure sensor (HSPPAD042A, Alps Alpine Co., Ltd.) that has a measurement range of 300 hPa to 1100 hPa. By applying the proposed method to such a smartphone with a relatively

small pressure measurement range, the estimation range can be expected to be able to be extended to a depth of about 20 m.

On the other hand, the proposed method has the disadvantage of low analysis granularity because water depth is estimated within the narrow measurement range of the barometric pressure sensor. Furthermore, in the proposed method, the amount of change in the deflection of the waterproof hard case decreases as the water depth increases, thus increasing the error at deeper water depths. Nevertheless, we expect that the proposed method, which allows the use of widely available smartphones as a simple replacement for dive computers, will make underwater activities more accessible.

5 CONCLUSION

In this study, we proposed a method to extend the limit value of the water depth estimation range with a smartphone from 2 m of the previous study method to around 20 m to 30 m by combining an inexpensive waterproof case with a decompression mechanism and machine learning technology. In addition, the evaluation of the depth estimation results at approximately 20 m depth showed that the RMSE was within 1 m. Therefore, we believe that the proposed method can serve as a simple and accessible alternative to dive computers. As we aim to further expand the range of smartphone use underwater, we also plan to research a model that estimates the change in internal case pressure that occurs when a smartphone display is touched underwater.

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